

Recall Boolean Algebra properties:

$$5a. \quad x \cdot 0 = 0$$

$$5b. \quad x + 1 = 1$$

$$6a. \quad x \cdot 1 = x$$

$$6b. \quad x + 0 = x$$

$$7a. \quad x \cdot x = x$$

$$7b. \quad x + x = x$$

$$8a. \quad x \cdot \bar{x} = 0$$

$$8b. \quad x + \bar{x} = 1$$

$$9. \quad \overline{\bar{x}} = x$$

$$10a. \quad x \cdot y = y \cdot x$$

Commutative

$$10b. \quad x + y = y + x$$

$$11a. \quad x \cdot (y \cdot z) = (x \cdot y) \cdot z$$

Associative

$$11b. \quad x + (y + z) = (x + y) + z$$

$$12a. \quad x \cdot (y + z) = x \cdot y + x \cdot z$$

Distributive

$$12b. \quad x + y \cdot z = (x + y) \cdot (x + z)$$

$$13a. \quad x + x \cdot y = x$$

Absorption

$$13b. \quad x \cdot (x + y) = x$$

$$14a. \quad x \cdot y + x \cdot \bar{y} = x$$

Combining

$$14b. \quad (x + y) \cdot (x + \bar{y}) = x$$

$$15a. \quad \overline{x \cdot y} = \bar{x} + \bar{y}$$

DeMorgan's theorem

$$15b. \quad \overline{x + y} = \bar{x} \cdot \bar{y}$$

$$16a. \quad x + \bar{x} \cdot y = x + y$$

$$16b. \quad x \cdot (\bar{x} + y) = x \cdot y$$

$$17a. \quad x \cdot y + y \cdot z + \bar{x} \cdot z = x \cdot y + \bar{x} \cdot z$$

Consensus

$$17b. \quad (x + y) \cdot (y + z) \cdot (\bar{x} + z) = (x + y) \cdot (\bar{x} + z)$$

We can prove the validity of these properties either by perfect induction or by performing algebraic manipulation.

Example 1: Let us prove the validity of the following logic equation

$$\underbrace{x_1 \cdot \bar{x}_3 + \bar{x}_2 \cdot \bar{x}_3 + x_1 x_3 + \bar{x}_2 x_3}_{LHS} = \underbrace{\bar{x}_1 \cdot \bar{x}_2 + x_1 x_2 + x_1 \bar{x}_2}_{RHS}$$

$$\begin{aligned} LHS &= x_1 \bar{x}_3 + \bar{x}_2 \bar{x}_3 + x_1 x_3 + \bar{x}_2 x_3 \\ &= x_1 \bar{x}_3 + \bar{x}_2 (\bar{x}_3 + x_3) + x_1 x_3 \\ &= x_1 \bar{x}_3 + \bar{x}_2 (1) + x_1 x_3 \\ &= \underbrace{x_1 \bar{x}_3 + \bar{x}_2}_{\text{}} + x_1 x_3 \\ &= x_1 (\bar{x}_3 + x_3) + \bar{x}_2 \\ &= x_1 (1) + \bar{x}_2 = x_1 + \bar{x}_2 \end{aligned}$$

$$\begin{aligned} RHS &= \bar{x}_1 \bar{x}_2 + x_1 x_2 + x_1 \bar{x}_2 \\ &= \bar{x}_1 \bar{x}_2 + x_1 (x_2 + \bar{x}_2) \\ &= \bar{x}_1 \bar{x}_2 + x_1 (1) \\ &= \bar{x}_1 \bar{x}_2 + x_1 \\ &= x_1 + \bar{x}_2 = LHS \end{aligned}$$

Example 2: Derive the **simplest** logical expression for the following function

$$\underline{F(A, B, C, D) = (A+B) \cdot (A+\bar{B}) \cdot (A+C) \cdot (A+D) \cdot B}$$

$$F = \underbrace{(A+B)(A+\bar{B})}_{\text{red bracket}} (A+C)(A+D)B$$

$$= [A + (B \cdot \bar{B})] (A+C)(A+D)B$$

$$= [A + 0] (A+C)(A+D)B$$

$$= \underbrace{A(A+C)}_{\text{red bracket}} (A+D)B$$

$$\downarrow$$
$$= \underbrace{A(A+D)}_{\text{red bracket}} B$$

$$= A \cdot B \quad \checkmark$$

We can write:

$$F = (A+B)(A+\bar{B})(A+C)(A+D)B = AB$$

Example 3: Prove the validity of the following

$$\underline{a.b + b.c + \bar{a}.c = a.b + \bar{a}.c}$$

$$LHS = ab + bc + \bar{a}c$$

$$= ab + 1.bc + \bar{a}c$$

$$= ab + (a + \bar{a}).bc + \bar{a}c$$

$$= \underline{ab} + \underline{abc} + \underline{\bar{a}bc} + \underline{\bar{a}c}$$

$$= ab(1+c) + \bar{a}c(b+1)$$

$$= ab(1) + \bar{a}c(1)$$

$$= ab + \bar{a}c = RHS$$

- You can solve the previous example in another way

$$LHS = ab + bc + \bar{a}c$$

$$= ab.1 + 1.bc + \bar{a}.1.c$$

$$= ab(c + \bar{c}) + (a + \bar{a})bc + \bar{a}(b + \bar{b})c$$

$$= \underbrace{abc + ab\bar{c}} + \underbrace{abc + \bar{a}bc} + \underbrace{\bar{a}bc + \bar{a}\bar{b}c}$$

$$= abc + ab\bar{c} + \bar{a}bc + \bar{a}\bar{b}c$$

$$RHS = ab + \bar{a}c$$

$$= ab.1 + \bar{a}.1.c$$

$$= ab(c + \bar{c}) + \bar{a}(b + \bar{b})c$$

$$= abc + ab\bar{c} + \bar{a}bc + \bar{a}\bar{b}c = LHS$$

It is valid

Example 4: Use algebraic manipulations to derive the simplest expression for the following function:

$$F(a, b, c) = (a+b+c)(a+b+\bar{c})(a+\bar{b}+c)(a+\bar{b}+\bar{c})(\bar{a}+b+c)(\bar{a}+b+\bar{c})(\bar{a}+\bar{b}+c)$$

